JPE Replication Report

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1 Paper Identification

ID	JPE-Richert-20240555
Author	Richert, Eric
Title	Quantity Commitments in Multiunit Auctions: Evidence from Credit Event Auctions
Iteration	3

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

In assessing compliance with our Data and Code Availability Policy, our reproducibility team has identified the following issues, which we ask you to address:

1. There are still numerous discrepancies in output. Your proposed fixes do not seem to have worked in all cases. Notice those are not small deviations but significantly different results.
2. Your computing environment description is still insufficient. Please improve.
3. The output of your in-text numbers scripts is different from what is printed in the paper. we see 209 vs 202 credit events, for example.

3 General

- [x] Data Availability and Provenance Statements
  - [x] Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - [ ] Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - [ ] License for Data (optional, but recommended)
  - [x] Details on each Data Source
- [x] Dataset list
- [x] Computational requirements

- [x] Software Requirements
- [x] Controlled Randomness (as necessary)
- [ ] Memory, Runtime, and Storage Requirements
- [x] Description of programs/code
  - [ ] License for Code (Optional, but recommended)
- [x] {REPLICATOR} to Replicators
  - [x] Details (as necessary)
- [x] List of tables and programs
- [x] References (Optional, but recommended)

4 High-level Description of Research Project

- [ ] uses observational data.
- [x] uses *confidential* observational data.
- [ ] uses data generated via experiment or trial by the authors (in field or lab).
  - [ ] For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
  - [ ] A PDF copy of the instructions given to participants in both original language and an English translation is included.
- [ ] uses data generated via survey by the authors (in field or online)
  - [ ] The programs used to administer the survey (e.g. qualtrics survey file) are included.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF of the original questionaire(s) and information on the selection and eligibility of participants is included.
- [ ] is a simulation study: the data are generated by an algorithm.
- [ ] is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

> {REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

> {REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years

> following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

> {REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
immspreads_clean.csv	no	yes	yes	yes	yes	yes
openinterest_clean.csv	no	yes	yes	yes	yes	yes
limitorders_clean.csv	no	yes	yes	yes	yes	yes
auctionlistid.csv	no	yes	yes	yes	yes	yes
bondtypes.csv	no	yes	yes	yes	yes	yes
gosyop23q6grk5y1.csv	no	yes	yes	yes	yes	yes

6 Requirements

- [x] README is in TXT, MD, or PDF format
- [x] README is accessible at the root of the package (not inside a folder)
- [x] Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- [x] Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- [ ] No requirements specified
- [x] Software Requirements specified as follows:
    - MATLAB R2023a
    - Toolboxes: Statistics and Machine Learning Toolbox, Optimization Toolbox, Parallel Computing Toolbox
- [ ] Computational Requirements specified as follows:
    - Hardware: 20-core cluster node or 3–4 days desktop
    - Storage: Sufficient space for large intermediate .mat files
    - Requires 56GB storage.

- [x] Time Requirements specified as follows:
    - approximately 20 hours on a 20-core cluster node, or 3–4 days on a desktop
- [x] Requirements are complete.

For missing requirements, see the list of required changes in Section 9

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals*. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 5 - Variables flagged in data: 6 - Code files with PII references: 38 - PII references in code: 585

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	`auctionlistid.csv`	1	name
Data	`bondtypes.csv`	2	name, loc
Data	`immspreads_clean.csv`	1	name
Data	`limitorders_clean.csv`	1	name
Data	`openinterest_clean.csv`	1	name
Code	`BsplineBasis3.m`	24	city, second, son, degree
Code	`BsplineEval3.m`	1	second
Code	`acrossrounds.m`	1	second
Code	`basic_estimator.m`	56	lon, lat, city
Code	`basic_estimator_copydrop.m`	56	lon, lat, city
Code	`bondpriceimport.m`	14	name, loc

File Type	File	Variables/References	PII Categories
Code	`bondprices.m`	1	lat
Code	`build_bondforestimation.m`	2	name
Code	`calculate_surplusraw.m`	5	lon, son
Code	`cdf_estimator.m`	1	lat
Code	`cdf_estimator_imm.m`	2	lat
Code	`complex_bootstrap.m`	6	second, lon, lat
Code	`complex_estimator_step1.m`	71	lon, lat, city
Code	`complex_estimator_step2.m`	12	lon, lat
Code	`complex_estimator_step2b.m`	11	lon, lat
Code	`data_summary.m`	27	name, block, loc, lat, second
Code	`doubleauctionOuter_1stepspecial_yin.m`	3	lat
Code	`dscatter.m`	11	loc, location, lon, name, second
Code	`emcdf.m`	1	lat
Code	`firststagebidding.m`	10	loc, lat
Code	`heatscatter.m`	9	name, lat
Code	`jacobianest.m`	13	loc, location, lat, second
Code	`main_cds.m`	3	lat, loc, location, name
Code	`normalize_prices.m`	28	lon, lat, second, name
Code	`npestimator.m`	20	lon
Code	`permn.m`	4	son, name, lat, second

File Type	File	Variables/References	PII Categories
Code	`postestimation_clean.m`	16	lat, second, son, lon, url
Code	`postmain_cfs.m`	1	lat
Code	`pricechangebs.m`	1	loc, location
Code	`robustnesschecks.m`	6	lat, lon
Code	`round1_quotescalibration.m`	8	lat, lon
Code	`smc_cfs_yin.m`	30	lat, block, loc
Code	`table2latex.m`	20	lat, name, son
Code	`truthfulpimmcheck.m`	3	lat
Code	`v_correction.m`	3	lon
Code	`weightedMedian.m`	4	lat
Code	`weightedcorrs.m`	100	lat, son, second, loc, location
Code	`weightsolnpricespartialgridINTs1_yin.m`	1	lon

See Appendix for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-06-08T14:18:44.890

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (`C:\` etc)
64	7	22	10	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/CDS/confidential-data-not-for-publication/gosyop23q6grk5y1.csv	129.31

7.2 Code description

There are provided 37 MATLAB, 1 of which is the master script (`code/main_cds.m`).

cloc	github.com/AlDanial/cloc v 2.06 T=0.10 s (724.0 files/s, 258898.1 lines/s)

Language	files	blank	comment	code
CSV	5	0	0	14416
MATLAB	64	1048	2041	7077
Markdown	1	144	0	304
——–	——–	——–	——–	——–
SUM:	70	1192	2041	21797

- [x] The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

Hardware:

- Nuvolos: Ubuntu 24.04.1 LTS, 16 vCPU + 16 GB RAM

Software:

- Matlab R2023a with Intel compilers

8 Replication

- Downloaded code and data from dropbox.
- Copied git repo and unzipped the replication package into this repo.
- Set up the computing environment on Nuvolos and scaled the hardware to 16 vCPU and 64 GB RAM.
- Modified `code/main_cds.m` to set `ncores = 16` to match the available cluster hardware as instructed in the `README.md`
- Set working directory and ran the code with MatLab 2023a as per `README.md`
- Committed changes

9 Findings

9.1 Data Preparation Code

- Program `main_cds.m` stopped after encountering error. Output not as expected.

9.2 Tables and Figures

Figure/Table #	Program	Line Number	Reproduced?
Figure 1A	normalize_prices.m	458	Yes
Figure 1B	normalize_prices.m	469	Yes
Figure 3	/		Yes
Figure 4 (left)	postestimation_clean.m	395	Yes
Figure 4 (middle)	postestimation_clean.m	408	Yes
Figure 4 (right)	postestimation_clean.m	416	Yes
Table 1	data_summary.m	130	Yes
Table 2	data_summary.m	158	Yes
Table 3	normalize_prices.m	263	Yes

Figure/Table #	Program	Line Number	Reproduced?
Table 4	postestimation_clean.m	219	Yes
Table 5	postmain_cfs.m	76	No
Table A.1	normalize_prices.m	437	Yes
Table OS.1	data_summary.m	77	No
Table OS.2	data_summary.m	305	Yes
Table OS.3	data_summary.m	200	Yes
Table OS.4	normalize_prices.m	326	Yes
Table OS.5	normalize_prices.m	379	Yes
Table OS.6	main_cds.m	153	No
Figure OS.3	bondprices.m	69	Yes
Figure OS.4	data_summary.m	356	Yes
Figure OS.5	data_summary.m	334	Yes
Figure OS.6	robustnesschecks.m	25, 31	Yes
Figure OS.7	firststagebidding.m	97	Yes
Figure OS.8	postestimation_clean.m	330	Yes

- `Table 2.tex` presents a small issue with the character & for lines 7 and 8.
- Table 5: estimates do not match

	Mean	Variance	$Cov(P^c, P^v)$	Surplus
Double Auction Price	[44.17,44.83]	[1186,1321]	[1241,1309]	[0.8,2.3]

Paper version

Table 5: Change in Auction Format

	Mean	Variance	$\mathrm{Cov}(P^c, P^v)$	Surplus
Double Auction Price	[44.18, 44.83]	[1173, 1328]	[1232, 1310]	[1.3, 4.1]

Replicated version

Figure 1: Table 5

- Table OS.1: estimates do not match

	Mean	Sd	[P10,P90]
N Bonds	10.77	27.70	[1,16.90]
Max maturity (years)	10.42	11.93	[0,26.9]
Min maturity (years)	2.97	4.92	[0,6.90]
Max coupon %	5.5	4.8	[0,10.74]
Min coupon %	2.7	3.3	[0,7.62]
FRN %	33.8	41.7	[0,100]

Paper version

Table 7: Eligible Bonds Description

	Mean	Sd	$[P_{10}, P_{90}]$
N Bonds	10.77	27.70	[1.00, 16.90]
Max maturity (years)	10.42	11.93	[2.00, 26.90]
Min maturity (years)	2.97	4.91	[0.00, 6.90]
Max coupon %	5.50	4.77	[0.00, 10.74]
Min coupon %	2.73	3.34	[0.00, 7.62]
FRN %	33.81	41.71	[0.00, 100.00]

Replicated version

Figure 2: Table OS.1

- Table OS.6: estimates do not match

the price limit is ±2 cents and ±4 cents.

	Double Auction	
	Supply=500	Supply=100
Mean Price $\mathbb{E}[\boldsymbol{P}\|P^M = 34.5]$	[35.99,36.27]	[36.87,36.88]
Std $(\boldsymbol{P}\|P^M = 34.5)$	[0.47,0.74]	[0.30,0.30]

Paper version

Table 12: Change in Auction Format (Bond Supply)

	Mean Price	SD Price	Surplus ($M)
Volume cap $= 500$	$[35.80, 36.14]$	$[0.45, 0.79]$	$[-0.9, 2.2]$
Volume cap $= 100$	$[36.26, 36.26]$	$[0.70, 0.70]$	$[-0.8, 1.9]$

Replicated version

Table OS.6

9.3 In-Text Numbers

In-text numbers	Program	Line Number	Reproduced?
Section 2 | Sample descriptives (N auctions, N bidders, NOI, etc.)	normalize_prices.m		No
Section 2 | Price cap statistic	firststagebidding.m		Yes
Section 6.1 | Status quo surplus bounds	calculate_surplusraw.m		No
Section 6.1 | Truthful bidding surplus bounds	postestimation_clean.m		No
Section 6.2 | Price bias (cents, %)	postestimation_clean.m		Yes
Section 6.3 | Risk SD, hedging effectiveness	postestimation_clean.m		Yes
Section 6.4 | Gain from full insurance	postestimation_clean.m		Yes
Section 7 | CF surplus, price bias, hedging	postmain_cfs.m		No
Appendix C.2 | Risk aversion comparison	riskaversion.m		Yes

In-text numbers	Program	Line Number	Reproduced?
Appendix C.5.1 | Truthful reporting incentives	truthfulpimmcheck.m		Yes
Appendix C.5.1 | Quote manipulation calibration	round1_quotescalibration.m		Yes
Appendix C.6 | Customer orders robustness	robustnesschecks.m		Yes
Appendix C.7 | Bid shading with positions	postestimation_clean.m		Yes
Appendix D | CF robustness (volume caps, positions)	smc_cfs_yin.m		No

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code
- For Section 2, again the log prints don't match one of the in-text numbers:
  - "There were 202 credit events [...] Of these, 47 were loan credit default swaps (LCDS) and 155 CDS."

```
========== SECTION 2: Sample Composition ==========
Total credit events: 209
LCDS auctions: 33
CDS auctions: 162
Fraction of bids at price floor/ceiling: 3.1%
Fraction of auctions clearing at cap: 15.0%
Fraction at cap conditional on nonzero NOI: 6.0%
```

- For Section 6.1, the log prints don't match one of the in-text numbers:
  - "The surplus of the status-quo design is $[-7, 48]M".

```
 Status quo surplus bounds: [$-8M, $47M]
========== SECTION 6.1: Surplus from Reallocation ==========
Truthful bidding surplus bounds: [$43M, $91M]
```

- For Section 7, the log prints don't match in-text numbers:

- "The counterfactual leads to total surplus $[0.8,2.3]M."

```
========== SECTION 7: Counterfactual Surplus ==========
Mean surplus ($M) [LB, UB]: [1.3, 4.1]
Max surplus ($M) [LB, UB]: [2.6, 7.9]
Min surplus ($M) [LB, UB]: [-0.1, 0.5]
```

- For Appendix D.3, the log prints don't match in-text numbers:
    - "leads to expected prices of [35.26,35.46] and standard deviations of [0.93,1.18]"

```
========== APPENDIX D.3: Counterfactual with Changes in Positions ==========
Expected price: [35.01, 35.23]
Std dev:        [1.36, 1.52]
```

9.4 Other Issues

- Missing hardware requirements. 'Desktop' is generic. Please specify cpu, ram etc.

10 Classification

- [ ] full reproduction
- [x] full reproduction with minor issues
- [ ] partial reproduction (see above)
- [ ] not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- [ ] None.
- [x] `Discrepancy in output` (either figures or numbers in tables or text differ)
- [ ] `Bugs in code` that were fixable by the replicator (but should be fixed in the final deposit)
- [ ] `Code missing`, in particular if it prevented the replicator from completing the reproducibility check
    - [ ] `Data preparation code missing` should be checked if the code missing seems to be data preparation code
- [ ] `Code not functional` is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- [ ] `Software not available to replicator` may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- [ ] `Insufficient time available to replicator` is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.

- [ ] `Data missing` is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- [ ] `Data not available` is marked when data requires additional access steps, for instance purchase or application procedure.
- [ ] `Missing README` is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements